

ENVIRONMENTAL DATA SCIENCE & ML

Air Quality Analysis & AQI Prediction Model

A machine learning approach for forecasting atmospheric pollutant levels and evaluating air pollution indicators.

Air Quality & Environmental Impact

Understanding Air Quality

Air quality is a measure of airborne pollutants in the environment, tracked continuously via standardized monitoring metrics like the Air Quality Index (AQI).

Escalating industrial activities, vehicle emissions, and energy demands lead to dangerous concentrations of toxic particulate matter and gaseous compounds.

Timely forecasting enables authorities to issue early public health warnings and institute preventative environmental controls.



Major Causes of Air Pollution



Fossil Fuel & Industry

Thermal power generation and manufacturing plants release sulfur dioxide (SO₂), carbon monoxide (CO), and hazardous volatile chemicals.



Vehicular Emissions

Internal combustion engines generate heavy tailpipe nitrogen dioxide (NO₂) and fine aerosols, forming urban smog.



Agriculture & Fires

Livestock operations, fertilizer usage, and agricultural burning release massive loads of ammonia (NH₃), methane, and ash.

Health & Ecosystem Risks

Critical Consequences

Fine particulate matter (PM_{2.5}) penetrates deep into human respiratory tracts, elevating risks of asthma, lung damage, and cardiovascular diseases.

Sulfur and nitrogen atmospheric compounds combine with rainwater to create acid rain, damaging soil micro-biomes and aquatic habitats.

Excess greenhouse gases trap thermal energy, driving global climate shifts and prolonged severe weather events.



Air Quality Index (AQI) Scale

AQI Range	Category	Health Impact & Severity
0 - 50	Good	Minimal health impact; air quality is considered satisfactory.
51 - 100	Satisfactory	Minor respiratory discomfort for sensitive individuals.
101 - 200	Moderate	Breathing discomfort to people with lung disease, asthma, or heart conditions.
201 - 300	Poor	Mild symptoms aggravate upon prolonged exposure for general population.
301 - 400	Very Poor	Respiratory illness risk for healthy population on prolonged exposure.
401 - 500	Severe	Serious impacts on healthy people and severe health effects on vulnerable groups.

Project Objectives & Scope

Core Objectives

Analyze multi-year historical pollutant datasets to uncover spatio-temporal trends and dependencies.

Construct machine learning regression and time-series pipelines to accurately predict AQI values.

Provide quantitative data insights to support environmental policymaking and public health warnings.

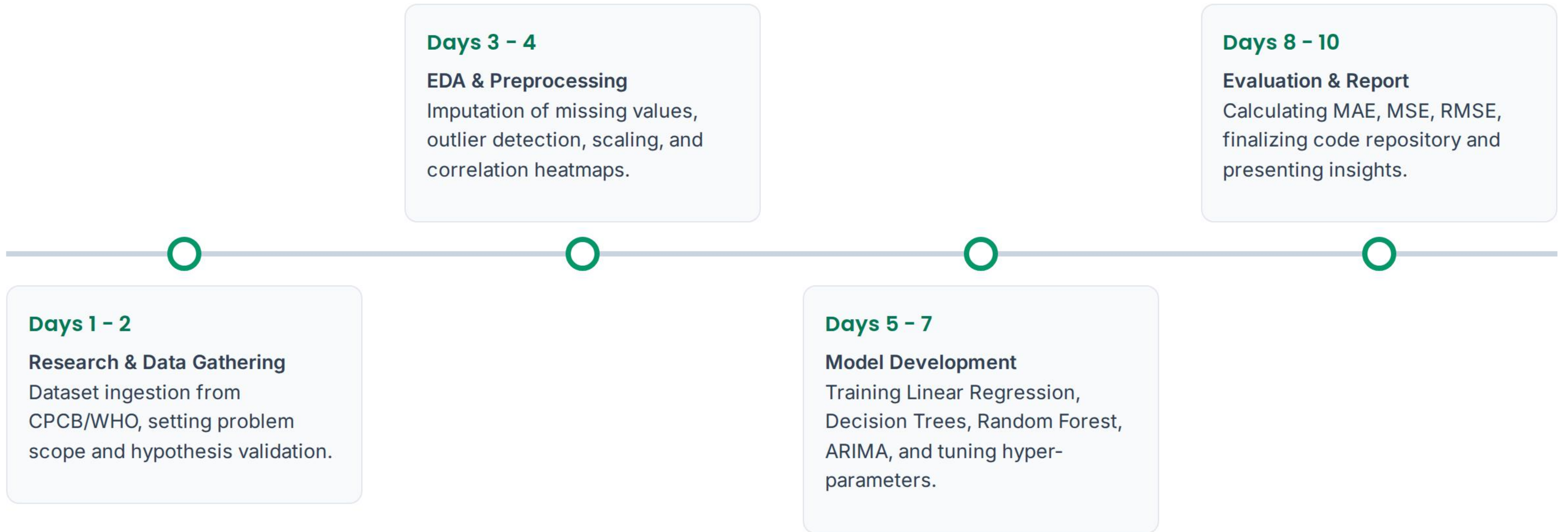
Project Scope

In-depth statistical study of criteria pollutants: PM_{2.5}, PM₁₀, NO₂, CO, SO₂, and O₃.

Evaluating feature correlation against meteorological variables like ambient temperature and humidity.

Deploying model benchmarks and open-source codebase for continuous monitoring adoption.

Accelerated 10-Day Plan



Dataset Feature Specifications

Feature Category	Attribute Name	Description & Measurement Unit
Temporal	Date / Timestamp	Date-time recording interval (hourly/daily aggregations)
Spatial	Location / Station ID	Monitoring station locality or city jurisdiction
Particulates	PM2.5, PM10	Fine and coarse particulate mass concentration (µg/m³)
Gaseous Pollutants	NO ₂ , CO, SO ₂ , O ₃	Chemical gases concentrations (µg/m³ or ppm)
Meteorology	Temp, Humidity, Wind	Ambient temperature (°C), relative humidity (%), wind velocity
Target Output	AQI	Calculated Air Quality Index numeric score (0–500)

System Architecture & Pipeline



1. Ingestion (`data/`)

Stores raw pollution monitoring records and outputs clean dataset files after validation.



2. Execution (`src/`)

Modular Python scripts for data cleaning, feature engineering, model training, and metrics reporting.



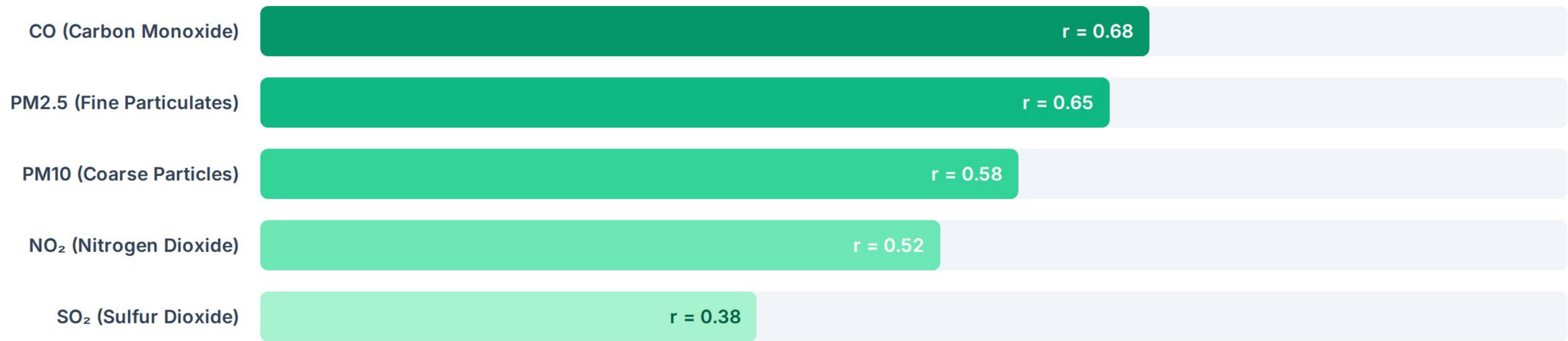
3. Exploration (`notebooks/`)

Interactive Jupyter notebooks executing exploratory statistical analysis, plots, and experiment logs.

Data Preprocessing Workflow

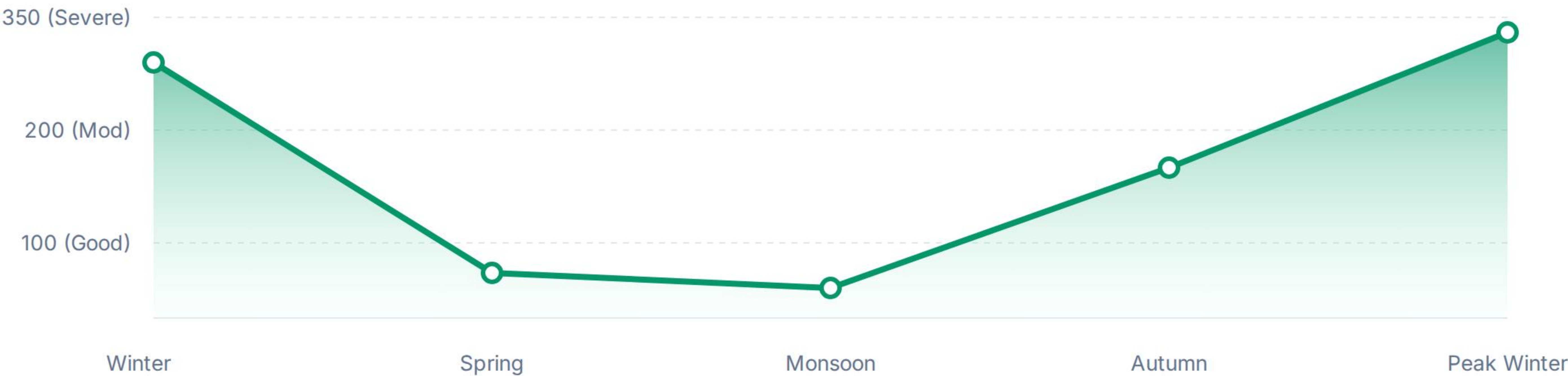
- ▼ **Missing Value Imputation:** Handled missing sensor data using median imputation and forward-fill for continuous time-series continuity.
- 📈 **Outlier Detection & Filtering:** Applied Interquartile Range (IQR) and Z-score trimming to eliminate faulty sensor spikes.
- ↔️ **Feature Scaling & Transformation:** Applied StandardScaler normalization to ensure equal weighting across disparate pollutant scales.
- ⚙️ **Feature Engineering:** Extracted seasonal indicators, lag parameters (t-1, t-2), and 7-day rolling window statistics for temporal models.

Pollutant Feature Correlations



Statistical correlation analysis indicates PM2.5 and CO exert the strongest direct linear effect on overall AQI readings.

Seasonal AQI Variations



Winter exhibits maximum pollution density due to atmospheric inversion, reduced wind speeds, and low temperatures.

Station & Regional Analysis



Urban vs. Rural Disparities

Dense urban industrial corridors show an average 2.4x higher particulate load compared to rural monitoring stations.

Automated telemetry stations (CPCB network) enable high-frequency continuous data stream collection.

Incorporating meteorological features significantly improves regional prediction accuracy across variable topography.

Model Development Framework

Linear Regression

Serves as the baseline linear modeling approach. Highly interpretable but limited in capturing non-linear feature interactions.

Decision Trees

Captures non-linear relationships effectively through recursive partitioning, but prone to overfitting without pruning.

Random Forest Regressor

Ensemble method utilizing bootstrap aggregation. Reduces variance drastically and yields high generalization accuracy.

ARIMA Modeling

Auto-Regressive Integrated Moving Average model specifically configured for short-term univariate time-series predictions.

Evaluation Metrics Benchmark

Algorithm	MAE (Mean Abs Error)	MSE (Mean Sq Error)	RMSE (Root Mean Sq Error)	R ² Score
Linear Regression	24.35	912.40	30.20	0.742
Decision Tree	14.80	420.15	20.50	0.881
Random Forest	8.12	145.60	12.06	0.958
ARIMA Time-Series	11.45	280.30	16.74	0.912

Random Forest outperformed traditional models across all regression metrics due to superior ensemble handling of complex non-linear data.

Model Accuracy Highlight

95.8%

Variance Explained (R^2 Score)

Key Experimental Findings

Random Forest Regressor provided the best overall performance, predicting unseen testing dataset AQI values with minimal prediction deviation (RMSE: 12.06).

Feature importance scoring confirmed PM2.5 and PM10 contribute over 60% weight towards overall AQI determination.

Team Roles & Coordination



Project Lead

Milestone planning, sprint management, documentation review, and team coordination.



Data Analyst

Dataset cleaning, missing data imputation, outlier filtering, and EDA visualizations.



ML Engineer

Model architecture development, hyperparameter tuning, metrics evaluation, and deployment scripts.

Risks & Mitigation Strategies

Identified Risk Factors

Sensor Inaccuracies: Missing or noise-corrupted telemetry data causing model prediction bias.

Overfitting: High-capacity decision trees memorizing localized atmospheric anomalies.

Mitigation Controls

Robust Pipeline: Automated Z-score filtering and median imputation during ingestion.

Cross-Validation: K-Fold cross-validation and hyperparameter regularization to ensure generalization.



Future Scope & Expansion

Integrating Deep Learning architectures (LSTM / GRU) for multi-step temporal forecasting.

Connecting real-time IoT air monitoring sensors and satellite telemetry feeds via automated APIs.

Deploying public web dashboards and mobile alert applications for real-time community safety warnings.

Questions & Discussion

Air Quality Analysis and Prediction Using Machine Learning

Thank you for your time and attention.

GitHub: github.com/air-quality-prediction | Project Documentation Included

Image Sources



https://img.freepik.com/premium-photo/urban-air-pollution_466739-19255.jpg

Source: www.magnific.com



<https://www.ppsthane.com/wp-content/uploads/2016/03/Air-quality-in-India-Industrial-Smoke.jpg>

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<https://www.opticalscientific.com/wp-content/uploads/2024/09/air-quality-sensors-02.jpg>

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Source: www.youtube.com